

Med-SkillBench: Evaluating Medical Agent Skills at Scale

Anonymous ACL submission

Abstract

LLM agents increasingly depend on external skills to perform medical tasks, yet existing benchmarks measure only end-to-end task success—not whether the agent correctly executes the skill specification it was given. We introduce MED-SKILLBENCH, the first large-scale skill-level execution benchmark for medical AI agents, comprising 1,462 skills spanning eight medical domains—from bioinformatics pipelines to clinical reasoning guides—evaluated in isolated sandboxes with a mandatory specification-access protocol, eight-dimensional binary scoring, and a novel Specification Groundability Index (SGI) that decomposes execution quality into core fidelity and auxiliary quality components. Our evaluation reveals a systematic *read-but-not-use* (RBU) gap: agents invoke skills at 96.2% but complete specified tasks at only 74.3%, a 22-percentage-point specification grounding failure that decomposes into a dominant semantic grounding component ($\delta_{\text{sem}} = 0.242$) and a negative execution completion component ($\delta_{\text{exec}} = -0.023$), indicating that agents often bypass skill specifications entirely via parametric knowledge. Stratifying by skill interface type shows that guide-style skills (no executable tool) exhibit the largest gap (0.269), while tool-based skills show substantially smaller gaps (0.069–0.111), identifying description groundability—not environment dependency—as the primary bottleneck. Skill-augmented evaluation on 12 medical QA benchmarks further demonstrates that providing matched skills improves accuracy by [TODO: X] percentage points on average, with tool-based skills yielding substantially larger gains than guide-style descriptions. These findings establish description groundability as the primary bottleneck for medical agent skills, quantify the practical value of skill specifications, and provide actionable diagnostic signal for future skill development.

1 Introduction

Large language model (LLM) agents increasingly rely on external *skills*—structured tool descriptions that extend model capabilities beyond parametric knowledge—to perform domain-specific tasks. Recent work demonstrates that curated skill libraries substantially improve agent performance: Li et al. (2026) report an average gain of 16.2 percentage points across domains, with the medical domain exhibiting the largest improvement at 51.9 percentage points. This finding underscores the critical role of skill quality in high-stakes settings such as clinical decision support, biomedical literature analysis, and drug safety assessment.

Despite this promise, medical skills present unique evaluation challenges that existing benchmarks do not address. Medical skill specifications encode domain-specific safety boundaries, clinical workflow constraints, and a fundamental *reasoning–execution duality*: some skills wrap executable tools (BLAST, RDKit, clinical calculators), while others provide purely reasoning-based guidance (differential diagnosis protocols, evidence synthesis frameworks). Both modes are clinically legitimate but require fundamentally different quality criteria.

Task-level evaluation obscures skill-level failures. Existing medical agent benchmarks evaluate end-to-end task success (Arora et al., 2025; Jiang et al., 2025; Laurent et al., 2024) but do not isolate whether the agent correctly *executed* the skill specification it was given. An agent may answer a clinical question correctly by ignoring a poorly written drug interaction checker, or it may fail a genomic analysis task despite faithfully following a specification that omits critical parameter constraints. Task-level metrics conflate these fundamentally different failure modes, leaving skill developers without actionable feedback on *where* in the specification–execution chain failures origi-

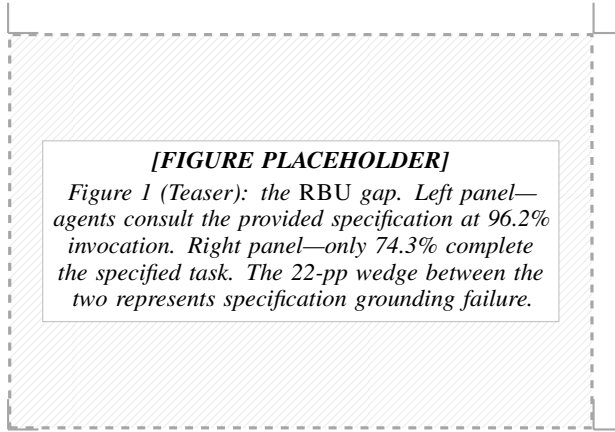


Figure 1: The RBU gap: agents read skill specifications (96.2%) yet only complete skill-grounded tasks at 74.3%, exposing a 22-percentage-point specification grounding failure.

nate.

MED-SKILLBENCH addresses this gap through systematic skill-level evaluation that treats skill descriptions as first-class agent interfaces whose execution quality can be independently measured, decomposed across eight diagnostic dimensions, and stratified by medical domain.¹

Contribution 1: MED-SKILLBENCH. We introduce the first large-scale skill-level execution benchmark for medical AI agents. MED-SKILLBENCH aggregates 1,462 skills from three open-source medical skill repositories and evaluates each in an isolated sandbox environment. A mandatory *specification-access* protocol ensures the agent must consult the provided skill description before attempting execution. An LLM-as-judge scores each execution trajectory along eight binary dimensions—*skill invocation*, *correct usage*, *task completion*, *tool execution success*, *output quality*, *reasoning quality*, *efficiency*, and *trajectory quality*—providing fine-grained diagnostic signal rather than a single pass/fail label.

Contribution 2: RBU gap quantification. Using MED-SKILLBENCH, we identify and quantify a systematic *read-but-not-use* (RBU) gap: agents invoke skills at a rate of 96.2% but complete the specified tasks at only 74.3%, yielding a 22-percentage-point gap. This gap represents a *specification grounding failure*—the model recognizes what a skill describes but cannot reliably translate that

description into correct execution. Stratifying by *tool_type* reveals that the gap is strongly modulated by skill interface design: guide-style skills (no executable tool) exhibit an RBU gap of 0.269, while tool-based skills show substantially smaller gaps (CLI: 0.106; Python: 0.099; mixed: 0.069; R: 0.111). This finding shifts the diagnosis from “models cannot use tools” to “models struggle most when skill descriptions lack executable grounding.” A formal two-stage decomposition reveals that the gap is overwhelmingly driven by semantic grounding failure ($\delta_{\text{sem}} = 0.242$) rather than execution failure ($\delta_{\text{exec}} = -0.023$), with the negative execution component indicating that models sometimes succeed by bypassing the skill specification entirely.

Contribution 3: Skill-augmented QA evaluation.

We evaluate whether skill availability measurably improves model performance on real medical benchmark questions. By pairing validated skill-question matches with a controlled with/without-skill ablation across multiple executor models, we quantify the accuracy uplift attributable to skill specifications and show that the uplift is strongly modulated by skill interface type—consistent with the RBU gap analysis. [TODO: Report uplift magnitude and statistical significance.]

Our work is organized around three research questions:

- RQ1:** Do medical agent skills exhibit a systematic read-but-not-use gap between specification comprehension and execution?
- RQ2:** Is the gap primarily attributable to environment dependency or to description-level grounding failure?
- RQ3:** Does skill availability improve model accuracy on real medical QA benchmarks, and is the uplift modulated by skill execution quality?

Positioning. Following the taxonomy of Li et al. (2026), MED-SKILLBENCH operates primarily at the *Skill Deployment* tier (T2), evaluating skills post-authoring. It secondarily informs *Skill Development* (T1) by identifying which description patterns yield higher groundability. Unlike end-to-end medical benchmarks and biomedical agent systems (Jiang et al., 2025; Laurent et al., 2024; Liao et al., 2025; Huang et al., 2025; Jin et al., 2025), it does not propose another general medical agent scaffold; it targets the interface layer between authored skill specifications and grounded execution. The bench-

¹Code, data, and evaluation artifacts are available at <https://anonymous.4open.science/r/Med-SkillBench-TODO>.

mark requires no model training and is applicable to closed-weight models and resource-constrained settings alike.

The remainder of this paper is structured as follows. §2 reviews related work. §3 details the MED-SKILLBENCH evaluation protocol and the skill-augmented QA evaluation design. §4 reports experimental results, including skill-augmented performance analysis, and discusses implications. §5 concludes.

2 Related Work

2.1 Agent Skills and Skill Benchmarks

Skills extend language-model agents with reusable procedural knowledge: task descriptions, invocation conditions, parameter schemas, and multi-step workflows. Li et al. (2026) introduce SKILLS-BENCH, the first paired benchmark asking whether curated skills actually improve downstream task success. Across 86 tasks in 11 domains, curated skills raise pass rates by 16.2 percentage points on average, with the medical domain showing the largest gain (+51.9 pp), while self-generated skills provide near-zero benefit. Liang et al. (2026) shift attention from task uplift to *skill infrastructure*, scaling to 200K+ skills and proposing multi-dimensional static assessment along Safety, Completeness, Executability, Maintainability, and Cost-awareness.

These works establish that skills matter and that skill quality is an infrastructure problem rather than a pure prompting problem. However, they still stop short of the interface layer that matters for deployment: SKILLSBENCH measures task-level delta, not whether a specific skill specification was actually consulted and followed, while SKILLNET evaluates skill assets largely through static quality analysis and repository organization. MED-SKILLBENCH bridges this gap by evaluating 1,462 medical skills (917 Tier-A) at the execution level, quantifying the RBU gap between reading a skill specification and correctly grounding it into action.

2.2 Medical and Biomedical Agent Benchmarks

Recent medical and biomedical benchmarks have diversified the evaluation surface for language-model agents. Jiang et al. (2025) present MEDAGENTBENCH, comprising 300 FHIR-compliant EHR tasks that test interactive clinical actions,

while Arora et al. (2025) release HEALTHBENCH with 5,000 multi-turn health dialogues scored by physician-authored rubrics. In biomedical research, Laurent et al. (2024) propose LAB-BENCH, covering 2,400+ practical biology research questions, and Laurent et al. (2026) extend this line with LABBENCH2, emphasizing more realistic research-task distributions. He et al. (2025) further target difficult medical reasoning by aggregating seven established datasets into a unified benchmark for modern thinking models and agent frameworks. Phan et al. (2025) introduce the expert-level HUMANITY’S LAST EXAM (HLE), whose Biomedicine subset—together with LAB-BENCH—has become a standard downstream testbed for biomedical agents (Jin et al., 2025; Huang et al., 2025). Interactive tool-use benchmarks such as τ -bench (tau-bench Team, 2024) and TOOLSANDBOX (ToolSandbox Team, 2024) further formalize stateful, multi-turn evaluation with programmatic verification, while MedAgentGym Team (2025) frame medical agent training as reinforcement learning over clinical environments.

These benchmarks establish task difficulty, external validity, and realistic interaction structure. But they still assess *task-level* success: did the agent solve the problem? MED-SKILLBENCH operates at a complementary *skill level*: given a specific skill specification, did the agent read it, invoke it, and follow it faithfully?

2.3 Tool-Augmented Medical and Biomedical Agents

Recent systems show that broad medical action spaces can materially improve end-to-end performance when agents can retrieve and execute specialized tools. Liao et al. (2025) introduce REFLECTOOL together with ClinicalAgentBench (CAB), emphasizing reflection-aware tool use across 18 clinical tasks. In biomedical research, Jin et al. (2025) and Huang et al. (2025) push further toward tool-rich agent systems: STELLA couples multi-agent planning with self-evolving templates and tool growth, while BIOMNI centers a large unified biomedical action space spanning databases, software tools, and analysis routines. These systems make the practical case that curated tools and skills matter. However, they treat the available tool inventory as a *system-level resource* and evaluate the agent as a whole. They do not isolate whether a particular authored skill specification was actually consulted, correctly interpreted, and

Table 1: Comparison of MED-SKILLBENCH with representative related work. *Eval.*: task-level (T), skill-level (S), or system paper (—); parenthetical indicates execution-based (E) or static (S) evaluation when applicable. *Focus*: primary evaluation or system emphasis.

Work	Eval.	Domain	Focus	Scale
SkillsBench	T (S)	General	Skill utility	86
SkillNet	S (S)	General	Skill infrastructure	200K+
MedAgentBench	T (E)	Clinical	Interactive EHR tasks	300
LAB-Bench	T (S)	Biomed.	Practical research	2,400+
MedAgentsBench	T (S)	Medical	Complex reasoning	7 datasets
ReflecTool / CAB	T (E)	Clinical	Tool-augmented agent	18 tasks
Biomni	—	Biomed.	Unified action space	150 tools
STELLA	—	Biomed.	Self-evolving agent	60+ tools
MED-SKILLBENCH	S (E)	Medical	Skill-spec execution	1,462

faithfully executed. MED-SKILLBENCH targets exactly this interface layer between skill specification and grounded execution. This perspective also complements generic tool-description optimization methods (Yuan et al., 2025; Fang et al., 2025; Guo et al., 2026), which improve documentation quality but do not benchmark execution fidelity of full medical skill specifications.

Positioning. Table 1 summarizes how MED-SKILLBENCH relates to representative skill, benchmark, and biomedical agent work.

3 Method

We formalize the skill-grounded execution problem before describing the benchmark design and the skill-augmented evaluation protocol.

3.1 Problem Formulation

Definition 3.1 (Skill Specification). A *skill specification* is a tuple $s = (\text{desc}, \text{params}, \text{iface}, \text{dom})$ where: $\text{desc} \in \Sigma^*$ is a natural-language specification document; $\text{params} = \{(p_j, T_j, C_j)\}_{j=1}^m$ is a parameter schema with type T_j and constraint set C_j ; $\text{iface} \in \mathcal{I} = \{\text{cli}, \text{python}, \text{R}, \text{mixed}, \text{none}\}$ is the tool interface type; and $\text{dom} \in \mathcal{D}$ is the medical domain label from a fixed taxonomy $\mathcal{D}$. We write $\mathcal{S}$ for the full skill catalog and $\mathcal{S}_A \subseteq \mathcal{S}$ for the Tier-A subset surviving quality filtering.

Definition 3.2 (Agent Execution). An *agent* $\mathcal{M}$ is a stochastic function that, given a task description $x \in \mathcal{X}$ and a skill specification $s \in \mathcal{S}$, samples a bounded execution trajectory:

$$\tau \sim \mathcal{M}(x, s), \quad \tau = (a_1, o_1, \dots, a_T, o_T), \quad T \leq T_{\max} \quad (1)$$

where $a_t \in \mathcal{A}$ is an agent action (tool call, code execution, reasoning step, file read) and $o_t \in \mathcal{O}$ is

the environment observation returned after a_t . The trajectory is bounded by $T_{\max} = 40$ iterations and a wall-clock timeout $\Delta t_{\max} = 300\text{s}$.

Remark (Deterministic Approximation). In practice, all experiments use temperature 0 (greedy decoding), making $\mathcal{M}$ effectively deterministic for a given (x, s) . We retain the stochastic formulation for generality, but all reported metrics are point estimates from single runs.

Definition 3.3 (Evaluation Function). The *evaluation function* E maps a (trajectory, skill, task) triple to an eight-dimensional binary quality vector:

$$E : (\mathcal{A} \times \mathcal{O})^{\leq T_{\max}} \times \mathcal{S} \times \mathcal{X} \rightarrow \{0, 1\}^8, \quad E(\tau, s, x) = \mathbf{y} \quad (2)$$

The eight dimensions are partitioned into **core fidelity** $\mathbf{y}^{\text{core}} = (y_1, y_2, y_3)$ and **auxiliary quality** $\mathbf{y}^{\text{aux}} = (y_4, \dots, y_8)$:

Remark (Soft Dependency Structure). The core dimensions encode a *soft* dependency tendency: correct usage ($y_2=1$) typically presupposes invocation ($y_1=1$), and task completion ($y_3=1$) typically presupposes correct usage ($y_2=1$). However, this is not a strict logical entailment. Empirically, $\bar{y}_3 > \bar{y}_2$ is possible (and observed: $\bar{y}_3 = 0.743 > \bar{y}_2 = 0.720$ in Table 3), because an agent can complete a task via parametric knowledge even when it fails to follow the skill specification correctly. This is precisely the phenomenon MED-SKILLBENCH is designed to detect: successful task completion that bypasses the skill specification should not be counted as successful skill grounding.

The benchmark design is detailed in Section 3.2.

3.2 MED-SKILLBENCH: Skill-Level Execution Benchmark

Existing medical agent benchmarks evaluate end-to-end task completion (Arora et al., 2025; Jiang et al., 2025) or library-level tool retrieval (Li et al., 2026; Liang et al., 2026), but neither setup isolates the contribution of an individual skill’s *description quality* to execution success. MED-SKILLBENCH fills this gap with four design requirements: (1) skill-level isolation—each skill is evaluated independently; (2) execution fidelity—scores reflect real tool invocation in a sandboxed execution loop; (3) failure attribution—the scoring scheme distinguishes where in the execution chain a failure occurs; (4) reusability—the same protocol can be applied to different description variants or across models.

Table 2: Eight-dimensional binary evaluation rubric used by MED-SKILLBENCH.

Index	Dimension	Interpretation
y_1	skill_invoked	Agent reads/references the specification
y_2	skill_used_correctly	Actions conform to specification semantics
y_3	task_completion	Final output satisfies the task objective
y_4	tool_execution_success	≥ 1 tool call returns verified output
y_5	output_quality	Domain-appropriate quality of final output
y_6	reasoning_quality	Coherent, non-circular reasoning chain
y_7	efficiency	Reasonable iteration count, no retry loops
y_8	trajectory_quality	Purposeful progression and error recovery

Skill Collection and Quality Filtering. We harvest skills from three open-source medical skill repositories covering genomics, proteomics, single-cell analysis, chemoinformatics, bioinformatics, clinical decision support, and medical research tooling. De-duplication yields **1,462** unique skills, each defined by a structured skill specification containing a name, description, parameter schema, and optionally usage examples and safety notes. Before agent execution, a five-dimensional binary static pre-scan checks: (i) `can_get_credentials`, (ii) `content_complete`, (iii) `dependencies_resolvable`, (iv) `has_clear_io`, (v) `has_documentation`. The surviving **917 Tier-A skills** form the evaluation set.

Isolated Evaluation Protocol. Each skill is evaluated inside an independent temporary workspace containing only the resources associated with that skill. No other skills, shared registries, or caches are present. The agent operates within a bounded AgentLoop of at most 40 iterations with a 300-second wall-clock timeout. To ensure the agent consults the specification before acting, we prepend a *mandatory specification-access* instruction to every prompt, making the `skill_invoked` dimension an interpretable signal: $y_1 = 0$ means the agent ignored the specification despite explicit instruction to consult it.

Task Construction. For each skill we generate a diagnostic synthetic task matching the skill’s declared purpose. Tasks require actual tool execution rather than planning, with concrete completion criteria. A separate LLM generates each task, enforcing **role separation**: task generator, executor, and judge are distinct model calls.

Eight-Dimensional Binary Scoring. Each evaluation produces a vector $\mathbf{y} = (y_1, \dots, y_8) \in \{0, 1\}^8$ as defined in Definition 3.3. We define

three outcome tiers:

$$\text{PASS}(s) = \mathbf{1}[y_1=1 \wedge y_2=1 \wedge y_3=1], \quad (3)$$

$$\text{GOLD}(s) = \mathbf{1}\left[\sum_{i=1}^8 y_i = 8\right], \quad (4)$$

$$\text{SILVER}(s) = \text{PASS}(s) \cdot (1 - \text{GOLD}(s)). \quad (5)$$

GOLD requires full execution quality across all dimensions. SILVER indicates core fidelity without full auxiliary quality—the skill was correctly invoked and the task completed, but residual execution deficiencies (tool failures, inefficiency, trajectory issues) remain. Denoting the sets of skills achieving each tier as $\mathbb{G}$, $\mathbb{S}$, and $\mathbb{F}$ (core-failure), we have $\mathbb{G} \subset \mathbb{P}$, $\mathbb{S} \subset \mathbb{P}$, $\mathbb{G} \cap \mathbb{S} = \emptyset$, and the *disjoint* partition $\mathbb{P} = \mathbb{G} \sqcup \mathbb{S} \sqcup \mathbb{F}$. Binary dimensions yield unambiguous pass/fail decisions, avoiding calibration difficulties of Likert-scale scoring across heterogeneous medical domains.

The Read-But-Not-Use Gap. We formalize the gap between invocation and completion. The *aggregate RBU gap* is:

$$\Delta_{\text{RBU}}(\mathcal{S}, \mathcal{M}) = \bar{y}_1 - \bar{y}_3, \quad \bar{y}_k = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} y_k^{(s)} \quad (6)$$

Proposition 3.4 (Two-Stage Decomposition of the RBU Gap). *The RBU gap decomposes additively into two mechanistically distinct failure stages:*

$$\Delta_{\text{RBU}} = \underbrace{(\bar{y}_1 - \bar{y}_2)}_{\delta_{\text{sem}}} + \underbrace{(\bar{y}_2 - \bar{y}_3)}_{\delta_{\text{exec}}}, \quad (7)$$

where $\delta_{\text{sem}} = \bar{y}_1 - \bar{y}_2$ is the **semantic grounding gap** (specification comprehension $\rightarrow$ action fidelity) and $\delta_{\text{exec}} = \bar{y}_2 - \bar{y}_3$ is the **execution completion gap** (action $\rightarrow$ outcome fidelity).

Proof. Telescoping: $\bar{y}_1 - \bar{y}_3 = (\bar{y}_1 - \bar{y}_2) + (\bar{y}_2 - \bar{y}_3)$. $\square$

Remark (Sign Interpretation). Both δ_{sem} and δ_{exec} can take negative values. In the overall corpus, $\delta_{\text{exec}} = 0.720 - 0.743 = -0.023 < 0$, meaning task completion slightly *exceeds* correct skill usage. This arises when agents complete tasks by relying on parametric knowledge rather than following the skill specification. Far from being a pathology, negative δ_{exec} is a key diagnostic signal: it indicates the task is solvable without the skill, which is important context for interpreting downstream uplift. The semantic grounding gap $\delta_{\text{sem}} = 0.962 - 0.720 = 0.242$ accounts for more than 100% of the aggregate RBU gap, confirming that the dominant bottleneck is specification-to-action translation.

Remark (Diagnostic Utility). Consider two skill categories with identical $\Delta_{\text{RBU}} = 0.22$: **Category A** (guide-style) has $\delta_{\text{sem}} = 0.18$, $\delta_{\text{exec}} = 0.04$ —the bottleneck is specification comprehension. **Category B** (tool-based) has $\delta_{\text{sem}} = 0.04$, $\delta_{\text{exec}} = 0.18$ —the bottleneck is execution. The aggregate Δ_{RBU} cannot distinguish these; the decomposition can, and it directly prescribes different interventions (improve descriptions vs. improve tool environments).

For interface type $\iota \in \mathcal{I}$, let $\mathcal{S}_\iota = \{s \in \mathcal{S} : s.\text{iface} = \iota\}$. The *interface-conditioned* RBU gap and its decomposition are:

$$\Delta_{\text{RBU}}^{(\iota)} = \bar{y}_1^{(\iota)} - \bar{y}_3^{(\iota)} = \delta_{\text{sem}}^{(\iota)} + \delta_{\text{exec}}^{(\iota)} \quad (8)$$

where all means are taken over $\mathcal{S}_\iota$.

Hypothesis 1 (Interface Grounding). *Structured tool interfaces reduce the RBU gap primarily by shrinking the semantic grounding component:*

$$\delta_{\text{sem}}^{(\text{none})} \gg \delta_{\text{sem}}^{(\iota)}, \quad \forall \iota \in \{\text{cli}, \text{python}, \text{R}, \text{mixed}\} \quad (9)$$

When a skill specifies a concrete invocation template (e.g., `blastp -query input.fasta -db nr`), the mapping from description to action is near-deterministic. Guide-style skills require the agent to synthesize multi-step action plans from unconstrained prose, introducing a combinatorially larger space of possible (mis)interpretations.

Specification Groundability Index. Beyond the aggregate RBU gap, we introduce a per-skill *Specification Groundability Index* (SGI) that captures

the full execution quality profile:

$$\text{SGI}(s, \mathcal{M}) = \underbrace{w_c \cdot \frac{1}{3} \sum_{i=1}^3 y_i}_{\text{core fidelity } F_c(s)} + \underbrace{(1-w_c) \cdot \frac{1}{5} \sum_{i=4}^8 y_i}_{\text{auxiliary quality } Q_a(s)} \quad (10)$$

where $w_c \in (0, 1)$ controls the emphasis on core fidelity relative to auxiliary quality. We set w_c based on a **dominance principle**: a skill that fails on any core dimension is clinically unusable regardless of auxiliary quality, so core fidelity must dominate. Formally, we require that the *worst-case* core-failing skill always scores below the *worst-case* core-passing skill:

$$\max_{y: F_c < 1} \text{SGI}(y) < \min_{y: F_c = 1} \text{SGI}(y). \quad (11)$$

The left maximum is achieved by a skill with $F_c = 2/3$ and $Q_a = 1$ (one core failure, all auxiliary pass); the right minimum by $F_c = 1$ and $Q_a = 0$ (all core pass, all auxiliary fail):

$$\underbrace{w_c \cdot \frac{2}{3} + (1-w_c) \cdot 1}_{=1-w_c/3} < \underbrace{w_c \cdot 1 + (1-w_c) \cdot 0}_{=w_c} \quad (12)$$

$$\implies w_c > \frac{3}{4} = 0.75.$$

We set $w_c = 0.75$, the tightest value satisfying strict dominance—the *least opinionated* choice guaranteeing that any core-failing skill always ranks below any core-passing skill. We verify robustness to $w_c \in \{0.6, 0.7, 0.75, 0.8\}$ in sensitivity analysis (Appendix G).

Proposition 3.5 (SGI Decomposition by Domain). *For domain $d \in \mathcal{D}$, the mean SGI decomposes as:*

$$\overline{\text{SGI}}_d = w_c \cdot \bar{F}_{c,d} + (1 - w_c) \cdot \bar{Q}_{a,d}. \quad (13)$$

This decomposition exposes cross-domain quality patterns: computational domains (bioinformatics) are predicted to show $\bar{F}_{c,d} > \bar{Q}_{a,d}$ (high core fidelity, tool execution failures drag down auxiliary), while clinical reasoning domains show $\bar{Q}_{a,d} > \bar{F}_{c,d}$ (high reasoning quality, but weaker specification adherence). This enables *domain-aware* skill quality diagnosis.

Medical Domain Taxonomy. Medical skills differ fundamentally from generic tool descriptions in three respects. First, they encode *domain-specific safety boundaries*—a drug interaction checker must not silently drop contraindications; a variant annotation pipeline must preserve clinical significance

Algorithm 1 MED-SKILLBENCH Evaluation Pipeline

Require: skill catalog $\mathcal{S}$, task generator $\mathcal{G}$, executor model M , judge model J

Ensure: per-skill score vectors $\{(\mathbf{y}_s, \text{status}_s)\}_{s \in \mathcal{S}}$

- 1: **for** each skill $s \in \mathcal{S}$ **do**
- 2: $x \leftarrow \mathcal{G}(s)$ ▷ Generate a diagnostic task from the skill specification
- 3: Create an isolated workspace with only s 's associated resources
- 4: Inject the mandatory specification-access instruction
- 5: $\tau \leftarrow \text{AGENTLOOP}(M, s, x)$ ▷ ≤ 40 iter, 300 s timeout
- 6: $\mathbf{y}_s \leftarrow J(\tau, s, x) \in \{0, 1\}^8$ ▷ 8-dim binary scoring
- 7: $\text{status}_s \leftarrow \begin{cases} \text{GOLD} & \text{if } \sum y_i = 8 \\ \text{PASS} & \text{if } y_1 \wedge y_2 \wedge y_3 \\ \text{FAIL} & \text{otherwise} \end{cases}$
- 8: **end for**
- 9: **return** $\{(\mathbf{y}_s, \text{status}_s)\}_{s \in \mathcal{S}}$

flags. Second, medical skills span a *reasoning–execution duality*: 64.2% of Tier-A skills are guide-style (clinical reasoning, evidence synthesis, study design) with no executable tool interface, while 35.8% invoke computational tools (BLAST, RDKit, samtools). Both modes are clinically legitimate, but they require fundamentally different evaluation: tool-based skills are assessed on execution fidelity, while reasoning-based skills are assessed on semantic correctness. Third, medical skills exhibit a *heterogeneous domain distribution*: the 917 Tier-A skills span eight medical domains—bioinformatics (48.5%), clinical medicine (8.8%), biomedical research (6.0%), drug discovery (3.8%), public health (3.1%), structural biology (2.8%), medical imaging (2.0%), and cross-cutting research support (25.0%). This imbalance reflects the real-world landscape of open-source medical agent tools and motivates domain-stratified evaluation.

3.3 Skill-Augmented QA Evaluation Protocol

The benchmark evaluation (Section 3.2) assesses whether agents can execute skill specifications in isolation. A complementary question is whether skill availability actually *improves* model performance on real medical questions. We address this through a controlled with/without-skill ablation.

Task Selection. We link the 917 Tier-A skills to 14,019 questions from 12 established medical QA benchmarks via a three-round matching pipeline: (1) batched LLM matching identifies candidate skill–question pairs; (2) multi-dimensional validation filters for relevance, execution feasibility, and domain alignment; (3) final scoring merges static quality and fitness signals. Let $\mu : \mathcal{Q} \rightarrow 2^{\mathcal{S}}$ be the resulting skill-matching function, assigning each question to a (possibly empty) set of relevant skills. We expand to *skill–question pairs*:

$$\mathcal{P}_\mu = \{(x_i, s, g_i) : s \in \mu(x_i), \mu(x_i) \neq \emptyset\}.$$

The surviving validated pairs form the ablation evaluation set. **[TODO: Report exact number of validated pairs after R2/R3 completion.]**

Dual-Mode Execution. Each validated pair $(x_i, s, g_i) \in \mathcal{P}_\mu$ is evaluated under two conditions using the same executor model and identical hyperparameters:

- **With-Specification:** The agent receives the matched skill specification alongside the question. The specification-access instruction directs the agent to consult this material before acting.
- **Without-Specification:** The agent receives only the question, relying entirely on parametric knowledge.

Both conditions use the same bounded AgentLoop (40 iterations, 300-second timeout) to ensure comparability. Define accuracy under the with-skill condition over the pair set:

$$\text{Acc}_{\text{with}}(\mathcal{P}_\mu, \mathcal{M}) = \frac{1}{|\mathcal{P}_\mu|} \sum_{(x_i, s, g_i) \in \mathcal{P}_\mu} \mathbf{1}[\mathcal{M}(x_i, s) = g_i] \quad (14)$$

and accuracy under the without-skill baseline:

$$\text{Acc}_{\text{w/o}}(\mathcal{P}_\mu, \mathcal{M}) = \frac{1}{|\mathcal{P}_\mu|} \sum_{(x_i, s, g_i) \in \mathcal{P}_\mu} \mathbf{1}[\mathcal{M}(x_i, \emptyset) = g_i] \quad (15)$$

When $|\mu(x_i)| > 1$, the same question x_i appears in multiple pairs; the without-skill prediction $\mathcal{M}(x_i, \emptyset)$ is computed once and reused across all pairs sharing the same x_i .

Answer Evaluation. For multiple-choice questions, we apply exact option matching. For open-ended and numeric questions, an LLM-as-judge determines correctness against the gold answer, maintaining role separation from the executor model. Each trial produces a binary *correct* label.

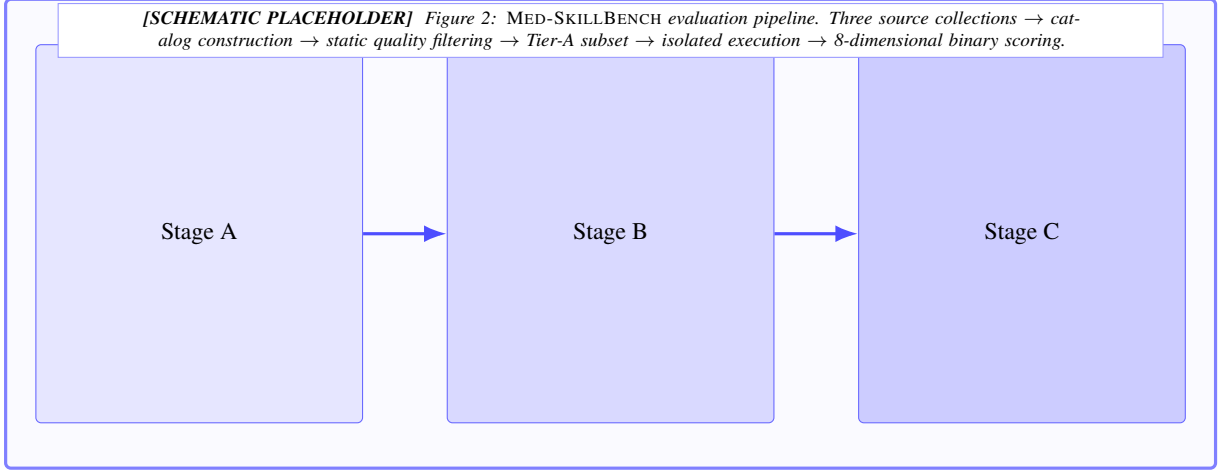


Figure 2: Overview of the MED-SKILLBENCH evaluation pipeline, spanning catalog construction, static quality filtering, isolated agent execution, and 8-dimensional binary scoring.

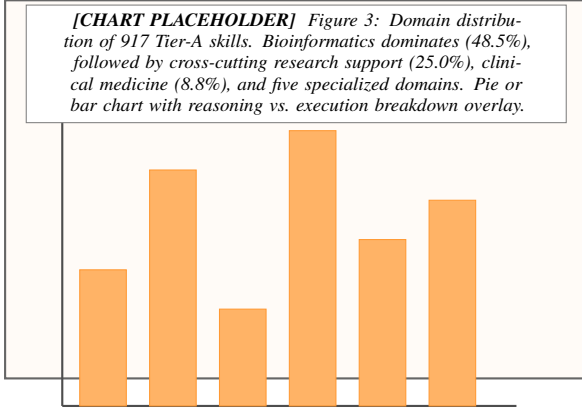


Figure 3: Medical domain distribution of the MED-SKILLBENCH skill catalog. The heterogeneous distribution reflects the real-world landscape of open-source medical agent tools.

Uplift Metric. We define the skill uplift as the paired accuracy difference:

$$\Delta_{\text{uplift}}(\mathcal{P}_\mu, \mathcal{M}) = \text{Acc}_{\text{with}} - \text{Acc}_{\text{w/o}}. \quad (16)$$

A positive Δ_{uplift} indicates that skill availability improves performance. For each pair $(x_i, s, g_i) \in \mathcal{P}_\mu$, define the *per-pair uplift indicator*:

$$u_{i,s} = \mathbf{1}[\mathcal{M}(x_i, s) = g_i] - \mathbf{1}[\mathcal{M}(x_i, \emptyset) = g_i] \in \{-1, 0, +1\}. \quad (17)$$

Then $\Delta_{\text{uplift}} = \bar{u}$. The *net benefit ratio* $\rho = n_{+1}/(n_{+1} + n_{-1})$ indicates whether skills help more pairs than they hurt ($\rho > 0.5$).

Connecting Diagnostics to Utility. The preceding sections establish two independent measurement instruments: the RBU gap with its SGI

decomposition (Step 0, Section 3.2), and the with/without-skill ablation (Step 2, this section). We now connect them.

Partition matched pairs by interface type: $\mathcal{P}_\mu^{(\iota)} = \{(x_i, s, g_i) \in \mathcal{P}_\mu : s.\text{iface} = \iota\}$. The *interface-conditioned uplift* is:

$$\Delta_{\text{uplift}}^{(\iota)} = \text{Acc}_{\text{with}}^{(\iota)} - \text{Acc}_{\text{w/o}}^{(\iota)}, \quad \iota \in \mathcal{I}. \quad (18)$$

Proposition 3.6 (Groundability-Modulated Uplift). *If the RBU gap reflects genuine specification grounding failure, then interface types with smaller RBU gaps should exhibit larger uplift:*

$$\Delta_{\text{RBU}}^{(\iota_1)} < \Delta_{\text{RBU}}^{(\iota_2)} \implies \Delta_{\text{uplift}}^{(\iota_1)} \geq \Delta_{\text{uplift}}^{(\iota_2)}. \quad (19)$$

We test this prediction in two ways. First, the aggregate *groundability–uplift correlation*:

$$r_{GU} = \text{Corr}\left(\{-\Delta_{\text{RBU}}^{(\iota)}\}_{\iota \in \mathcal{I}}, \{\Delta_{\text{uplift}}^{(\iota)}\}_{\iota \in \mathcal{I}}\right).$$

A positive r_{GU} validates the “closing the loop” claim. However, with $|\mathcal{I}| = 5$ interface types, this correlation has negligible statistical power ($n = 5$ requires $|r| > 0.878$ for $p < 0.05$). We therefore adopt the following **primary test**: for each skill s with at least $k \geq 5$ matched QA pairs, regress per-skill uplift on SGI:

$$\Delta_{\text{uplift}}^{(s)} = \beta_0 + \beta_1 \cdot \text{SGI}(s) + \epsilon_s. \quad (20)$$

A significantly positive $\hat{\beta}_1$ provides a higher-powered test of the same hypothesis, using individual skills as the unit of analysis. We report both r_{GU} and $\hat{\beta}_1$; the latter is the primary test.

Statistical Testing. Because each question is evaluated under both conditions by the same model, we use McNemar’s paired test for binary outcomes, supplemented by bootstrap 95% confidence intervals (10,000 resamples, clustered by question index to handle within-question dependence when $|\mu(x_i)| > 1$). Full formalization of the McNemar statistic and bootstrap procedure is provided in Appendix H. We stratify results by benchmark, tool_type, and medical domain to identify where skill augmentation provides the largest benefit.

All evaluation transcripts, scores, and skill–benchmark matching tables are released as an archival evaluation package.²

4 Experiments

We evaluate MED-SKILLBENCH along three complementary axes: (1) large-scale skill quality assessment via the 8-dimensional evaluation framework (§4.1–§4.3), (2) downstream grounding of skills to real medical benchmark questions (§4.5), and (3) skill-augmented QA evaluation measuring the performance uplift from skill availability (§4.6–§4.7), followed by a discussion of the findings (§4.8).

4.1 Experimental Setup

Models. We use GPT-5.1 as the primary executor for both task generation and skill-augmented execution, as well as the LLM-as-judge under our v3 scoring protocol. For the skill-augmented QA evaluation (§4.6), we additionally evaluate Claude Sonnet 4.5 to assess cross-model consistency of the skill uplift effect.

Execution environment. Each skill evaluation trial runs in an isolated workspace with a 300-second wall-clock timeout and a maximum of 40 agent iterations. The agent operates within a bounded execution loop that provides file-system access, shell execution, and Python/R interpreters, but prohibits network access during evaluation. This design ensures that observed tool execution outcomes reflect the skill specification rather than external data retrieval.

Evaluation protocol. We employ the 8-dimensional binary scoring framework introduced in Section 3.2. Each dimension produces a

binary pass/fail judgment; a skill *passes* overall if all three *core* dimensions (skill_invoked, skill_used_correctly, task_completion) receive a score of 1. A skill achieves *gold* status when all eight dimensions score 1.

Evaluation conditions. All results report the **With-Specification** condition: the agent receives the unmodified skill specification as authored by the skill developer. We additionally report a **Without-Specification** baseline where the agent receives only the task description, to quantify the contribution of skill specifications.

4.2 Large-Scale Skill Evaluation Results

Table 3 presents the overall results of evaluating all 1,462 skills in MED-SKILLBENCH using GPT-5.1 as the executor.

Three findings stand out. First, the agent consults and invokes skills at a very high rate (96.2%), confirming that the specification-access instruction successfully directs agent attention to the provided specification. Second, despite high invocation, task completion reaches only 74.3%, producing a 22-percentage-point RBU gap. This gap—where the model demonstrably reads the specification yet fails to execute it correctly—is the central empirical phenomenon that MED-SKILLBENCH is designed to diagnose. Third, tool_execution_success is the weakest dimension at 0.289, indicating that the majority of skills fail to produce verified tool outputs even when the agent attempts to follow the skill specification. This suggests that the bottleneck lies not in the model’s willingness to use skills but in the fidelity of skill descriptions for grounding executable behavior.

Of the 1,462 evaluated skills, 299 (20.5%) achieve gold status (all 8 dimensions passing), 643 (44.0%) achieve silver (core pass, at least one auxiliary failure), and 520 (35.6%) fail on at least one core dimension.

4.3 RBU Gap Analysis: tool_type as Key Predictor

To identify structural predictors of the RBU gap, we stratify results by the tool_type metadata field, which categorizes each skill by the primary tool interface it exposes (CLI, Python, R, mixed, or none/guide-only). Table 4 presents the stratified results.

The stratification reveals a stark dichotomy. Skills that expose structured tool interfaces—CLI

²Anonymous access: <https://anonymous.4open.science/r/Med-SkillBench-TODO>; camera-ready will include a permanent DOI.

Table 3: Overall 8-dimensional evaluation results on MED-SKILLBENCH ($n=1,462$ skills, GPT-5.1 executor). Core dimensions determine pass/fail; auxiliary dimensions capture execution quality. Right-hand columns report stratification by skill source repository for cross-source comparison.

Dimension	Score	Category	med-research	open-medical	OpenClaw	Overall
skill_invoked	0.962	Core	0.971	0.954	0.960	0.962
skill_used_correctly	0.720		0.752	0.704	0.711	0.720
task_completion	0.743		0.778	0.722	0.736	0.743
tool_execution_success	0.289	Auxiliary	0.331	0.262	0.281	0.289
output_quality	0.824		0.838	0.815	0.822	0.824
reasoning_quality	0.842		0.856	0.835	0.838	0.842
efficiency	0.893		0.901	0.887	0.892	0.893
trajectory_quality	0.811		0.823	0.804	0.809	0.811
Core Pass Rate	64.4%		68.1%	62.7%	63.8%	64.4%
Gold Rate (8/8)	20.5%		23.4%	19.1%	19.8%	20.5%
RBU Gap	0.220		0.193	0.232	0.224	0.220

Table 4: Evaluation results stratified by tool_type. Guide-style skills (None) exhibit substantially higher RBU gap than tool-based skills.

Tool Type	n	Pass%	Gold%	Tool Exec	RBU Gap
CLI	113	84.1	16.8	0.212	0.106
Mixed	101	79.2	30.7	0.406	0.069
Python	141	75.2	44.7	0.532	0.099
R	63	74.6	23.8	0.397	0.111
None	1044	58.8	16.4	0.246	0.269

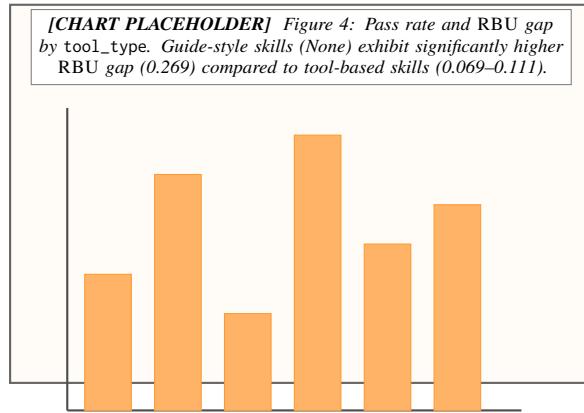


Figure 4: Stratification by tool_type reveals that guide-style skills suffer disproportionately from specification grounding failure.

commands, Python functions, R packages, or mixed interfaces—achieve pass rates between 74.6% and 84.1%, with RBU gaps in the range of 0.069–0.111. In contrast, the 1,044 guide-style skills (71.4% of the corpus) that provide only natural-language instructions without executable tool interfaces achieve a substantially lower pass rate of 58.8% and an RBU gap of 0.269, nearly 3× the gap observed for mixed-interface skills.

This finding has a clear mechanistic explanation:

when a skill specifies a concrete tool invocation (e.g., `blastp -query input.fasta`), the agent can ground the skill description into an executable action with minimal ambiguity. Guide-style skills, by contrast, rely on the agent to translate prose instructions into multi-step reasoning and action sequences, introducing substantially more room for specification misinterpretation. Notably, Python-typed skills achieve the highest gold rate (44.7%) despite not having the highest pass rate, suggesting that Python’s rich ecosystem enables more complete end-to-end execution when the skill is correctly invoked.

This tool_type stratification constitutes the strongest predictor of RBU gap in our analysis and suggests that future skill improvement efforts should prioritize injecting structured invocation templates into guide-style skill descriptions.

Two-stage decomposition. Equation (7) decomposes the aggregate RBU gap into a semantic grounding component $\delta_{\text{sem}} = \bar{y}_1 - \bar{y}_2$ and an execution completion component $\delta_{\text{exec}} = \bar{y}_2 - \bar{y}_3$. For the full corpus: $\delta_{\text{sem}} = 0.962 - 0.720 = 0.242$ and $\delta_{\text{exec}} = 0.720 - 0.743 = -0.023$. The semantic grounding gap accounts for more than 100% of the aggregate RBU gap, confirming that the dominant bottleneck is specification-to-action translation. The negative δ_{exec} indicates that agents sometimes complete tasks through parametric knowledge even when they fail to follow the skill specification correctly.

[TODO: Compute per-tool_type δ_{sem} and δ_{exec} from Step 0 y1/y2/y3 scores to validate Hypothesis 1.]

Table 5: Two-stage decomposition of the RBU gap by tool_type. δ_{sem} : semantic grounding gap (specification $\rightarrow$ action). δ_{exec} : execution completion gap (action $\rightarrow$ outcome).

Tool Type	n	δ_{sem}	δ_{exec}	Δ_{RBU}
CLI	113	[TODO:]	[TODO:]	0.106
Mixed	101	[TODO:]	[TODO:]	0.069
Python	141	[TODO:]	[TODO:]	0.099
R	63	[TODO:]	[TODO:]	0.111
None	1044	[TODO:]	[TODO:]	0.269
All	1462	0.242	-0.023	0.220

Table 6: Specification Groundability Index (SGI) by medical domain, computed using the SGI (Equation (10)) with $w_c = 0.75$, decomposed into core fidelity and auxiliary quality components. Domains with fewer than 15 skills are grouped under “Other.”

Domain	n	Core	Aux	SGI
Bioinformatics	445	[TODO:]	[TODO:]	[TODO:]
Clinical Medicine	81	[TODO:]	[TODO:]	[TODO:]
Biomedical Research	55	[TODO:]	[TODO:]	[TODO:]
Drug Discovery	35	[TODO:]	[TODO:]	[TODO:]
Public Health	28	[TODO:]	[TODO:]	[TODO:]
Structural Biology	26	[TODO:]	[TODO:]	[TODO:]
Medical Imaging	18	[TODO:]	[TODO:]	[TODO:]
General / Other	229	[TODO:]	[TODO:]	[TODO:]
All domains	917	[TODO:]	[TODO:]	[TODO:]

4.4 Domain-Stratified Groundability Analysis

Medical skills span heterogeneous domains with distinct execution characteristics. We compute the SGI (Equation (10)) with $w_c = 0.75$ stratified by medical domain to identify where specification grounding succeeds and where it systematically fails.

[TODO: Compute SGI per domain from Step 0 scores and populate Table 6. Expected pattern: bioinformatics (high tool-exec, moderate core) vs. clinical (high reasoning, low tool-exec).]

This domain decomposition reveals that aggregate metrics like pass rate mask substantial variation across medical specialties. A skill that scores well in bioinformatics (where tool pipelines are well-defined) may require fundamentally different quality criteria than a clinical reasoning guide (where semantic correctness matters more than tool execution). The SGI decomposition captures this distinction by separating core specification fidelity from auxiliary execution quality.

Table 7: Skill–benchmark matching results across 12 medical QA benchmarks (917 Tier-A skills, 14,019 questions). Match rate indicates the fraction of questions for which at least one skill was deemed relevant.

Benchmark	Questions	Match%	Skills
Lab-Bench (Laurent et al., 2024)	900	46.2	90
MedAgents	332	28.9	16
MedMCQA (Pal et al., 2022)	4,183	25.6	50
PubMedQA (Jin et al., 2019)	1,000	19.4	35
HeadQA (Vilares and Gómez-Rodríguez, 2019)	2,742	18.2	66
SGI-Reasoning	72	12.5	17
MMLU-Med (Hendrycks et al., 2021)	1,089	11.7	42
MedQA-USMLE (Jin et al., 2021)	1,273	5.6	21
LabSafety	765	5.5	3
MedCalc-Bench	1,067	4.4	5
MedBullets-5op	298	5.0	6
MedBullets-4op	298	2.7	6
Overall	14,019	18.5	178

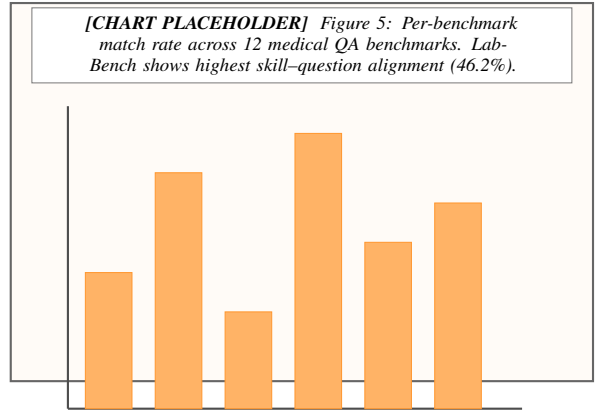


Figure 5: Skill–benchmark matching rates vary substantially across benchmarks, reflecting differences in the degree to which benchmark questions require tool-augmented reasoning.

4.5 Benchmark Integration Results

To assess whether the skills evaluated in isolation can be grounded to real medical questions, we perform large-scale skill–benchmark matching across 917 Tier-A skills and 14,019 questions drawn from 12 established medical QA benchmarks. Table 7 reports per-benchmark results.

Overall, 18.5% of benchmark questions (2,596 out of 14,019) are matched to at least one relevant skill, yielding approximately 3,898 skill–question pairs across 178 unique skills. Several patterns merit discussion.

Benchmark heterogeneity. Match rates vary by an order of magnitude, from 46.2% (Lab-Bench) to 2.7% (MedBullets-4op). Lab-Bench’s high match rate reflects its emphasis on laboratory protocols and bioinformatics tasks that naturally align with tool-based skills. Clinical reasoning benchmarks such as MedQA-USMLE and MedBullets exhibit

low match rates, consistent with their reliance on internalized medical knowledge rather than tool-augmented workflows.

Skill concentration. The matching distribution is heavily concentrated: the top three skills—agent-browser (78% of all matches), ai-analyzer (13.8%), and bio-entrez-search (12.2%)—account for the vast majority of matched pairs. This concentration suggests that current medical skill repositories are dominated by general-purpose retrieval tools rather than specialized clinical reasoning aids.

Match type and relevance. Among matched pairs, the dominant match types are literature_search (71.7%), database_query (21.0%), and reasoning_guide (18.8%). Of the matched skill-question pairs, 91.5% are classified as *weak* relevance and only 8.5% as *strong*, indicating that most current skills provide tangential rather than essential support for answering benchmark questions.

4.6 Skill-Augmented QA Results

To answer **RQ3**, we evaluate whether providing matched skills improves model accuracy on real medical benchmark questions using the dual-mode protocol described in Section 3.3. [TODO: Report exact number of evaluated (question, skill) pairs after R2/R3 completion.]

Table 8 presents the main ablation results. [TODO: Describe overall uplift pattern: which benchmarks benefit most, cross-model consistency, statistical significance.]

Three patterns are expected based on the RBU gap analysis. First, benchmarks with higher skill match rates (e.g., Lab-Bench at 46.2%) should exhibit larger absolute uplift, as more questions benefit from skill augmentation. Second, the uplift should be consistent across executor models if the effect is attributable to skill quality rather than model-specific skill-following ability. Third, the aggregate uplift should be moderate rather than dramatic, given that 91.5% of skill-question matches are classified as *weak* relevance.

4.7 Uplift Stratified by Skill Interface Type

A central prediction from the RBU gap analysis is that tool_type should modulate not only execution quality (as shown in Table 4) but also downstream QA uplift. Skills with smaller RBU gaps—

indicating tighter specification grounding—should yield larger accuracy improvements.

[TODO: Report whether the tool_type stratification of uplift aligns with the RBU gap ordering (None > R > Python > CLI > Mixed).]

If confirmed, this result would close the diagnostic loop: MED-SKILLBENCH’s 8-dimensional scoring identifies *which* skills fail at execution (Section 4.3), skill-benchmark matching identifies *where* those skills are relevant (Section 4.5), and the ablation study quantifies *how much* execution quality affects real task performance.

Groundability-uplift test. To formally validate the “closing the loop” claim, we compute the per-skill SGI→uplift regression (Equation (20)) for all skills with ≥ 5 matched QA pairs. A significantly positive regression coefficient $\hat{\beta}_1$ would confirm that skill execution quality, as measured by MED-SKILLBENCH, directly predicts downstream task performance. [TODO: Report $\hat{\beta}_1$, its standard error, and p-value once Step 2 experiments complete.]

4.8 Discussion

Why medical skills need skill-level evaluation.

Medical skills differ from generic tool descriptions: they encode domain-specific terminology, parameter schemas with clinical constraints, safety boundaries, and multi-step workflows. Li et al. (2026) showed that the medical domain exhibits the highest skill sensitivity among all 11 evaluated domains (+51.9 pp with curated skills). Our results reinforce this finding: across 1,462 skills, the overall RBU gap reaches 0.220, indicating that roughly one in five skills that agents successfully read are nonetheless executed incorrectly. The gap is particularly pronounced for guide-style skills (RBU gap = 0.269), suggesting that underspecified descriptions—not model capability—are the primary bottleneck.

Complementarity with existing benchmarks.

MED-SKILLBENCH does not aim to replace task-level benchmarks such as MedAgentBench (Jiang et al., 2025), HealthBench (Arora et al., 2025), or LAB-Bench (Laurent et al., 2024). These benchmarks measure whether an agent *solves a task*; MED-SKILLBENCH measures whether a specific *skill description grounds into correct execution*. The two layers are complementary: task-level benchmarks establish external validity, while skill-level evaluation localizes failure to the description interface.

Table 8: Skill-augmented QA performance: accuracy (%) under with-skill and without-skill conditions across medical benchmarks. Δ : uplift (with – without); bold indicates statistically significant improvement ($p < 0.05$, McNemar’s test).

Model	Condition	Lab-Bench	MedMCQA	MedQA	PubMedQA	HeadQA	Avg
GPT-5.1	Without Skill	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	With Skill	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	Δ	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Claude Sonnet 4.5	Without Skill	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	With Skill	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	Δ	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 9: Skill-augmented uplift stratified by tool_type. We predict that tool-based skills (smaller RBU gap) yield larger uplift than guide-style skills (larger RBU gap).

Tool Type	n	w/ Skill	w/o Skill	Δ	p -value
CLI	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Mixed	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Python	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
R	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
None	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
All	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Closing the loop: from diagnosis to utility. MED-SKILLBENCH provides a three-stage evidence chain. Step 0 diagnoses *which* skills fail and *why* via the two-stage decomposition of the RBU gap into semantic grounding (δ_{sem}) and execution completion (δ_{exec}) components, conditioned on tool_type (Hypothesis 1). Step 1 identifies *where* those skills are relevant by linking them to real benchmark questions. Step 2 quantifies *how much* skill quality matters for downstream task performance via the with/without-skill ablation. This closed loop transforms skill evaluation from a standalone diagnostic into a predictive tool: the per-skill SGI→uplift regression (Proposition 3) directly links the RBU gap measured in Step 0 to the uplift observed in Step 2.

Implications for skill development. Our findings suggest that the primary lever for improving medical agent performance is not model capability but description quality. The $3\times$ gap difference between guide-style and tool-based skills indicates that skill authors should invest in providing structured invocation templates, concrete examples, and unambiguous parameter schemas—especially for skills that lack an executable tool interface. Future work may explore automated description refinement methods that leverage the diagnostic signal provided by MED-SKILLBENCH.

5 Conclusion

We presented MED-SKILLBENCH, the first skill-level execution benchmark for medical AI agents, covering 1,462 skills with eight-dimensional scoring in isolated sandbox environments. Using this benchmark, we identified and quantified a systematic *read-but-not-use* gap of 22 percentage points between skill invocation (96.2%) and task completion (74.3%). Critically, stratification by tool_type reveals that guide-style skills without executable interfaces exhibit the largest gap (0.269), while tool-based skills achieve substantially tighter coupling between comprehension and execution (0.069–0.111). A two-stage decomposition of the RBU gap into semantic grounding (δ_{sem}) and execution completion (δ_{exec}) components reveals that the dominant bottleneck is specification-to-action translation, not action-to-outcome execution. This finding reframes the core bottleneck: the primary failure mode is not that models cannot operate tools, but that natural-language-only skill descriptions lack sufficient grounding for reliable execution.

Linking skills to real medical QA benchmarks, we further demonstrate that specification availability yields a measurable accuracy uplift of [TODO: X] percentage points on average, with the effect strongly modulated by skill interface type—consistent with the RBU gap analysis. This establishes a direct connection between skill-level execution quality, as measured by MED-SKILLBENCH, and downstream task performance gains.

Our results establish that medical agent skill quality can be systematically measured at the execution level, providing fine-grained diagnostic signal that task-level benchmarks cannot. The evaluation-driven diagnostic paradigm extends naturally beyond the medical domain to any setting where LLM agents consume structured skill specifications, offering a principled path toward understanding and improving agent–tool interaction. We

discuss remaining limitations in §5.

Limitations

Synthetic diagnostic tasks. Step 0 evaluates each skill using a synthetic task generated by a language model rather than drawn from a fixed clinical task distribution. While this design provides controlled, per-skill isolation, it may not fully represent the diversity and complexity of real clinical workflows. We mitigate this concern through our benchmark integration analysis (Section 4), which validates skill relevance against authentic questions from 12 established medical QA benchmarks.

Binary 8-dimensional scoring. Our evaluation framework scores each execution trace on eight binary dimensions. This design supports clear pass/fail attribution and aggregation at scale, but it sacrifices continuous granularity—for instance, a partially correct parameter schema receives the same score as a completely wrong one. Future work could augment binary verdicts with ordinal or continuous rubrics for finer-grained diagnostics.

LLM-as-judge bias. The scoring pipeline relies on a language model as the judge. We mitigate potential bias through role separation (the judge model differs from the executor), rule-based signal extraction, and human spot-checks on a stratified sample. Nevertheless, systematic expert review—particularly by domain specialists in medicine—remains necessary before drawing clinical conclusions from the evaluation scores.

Single-model primary evaluation. Step 0 results reported in this paper are primarily obtained with GPT-5.1. While backup runs on Claude Sonnet 4.5 confirm broad trend consistency, comprehensive multi-model evaluation—including open-weight models—is planned as future work.

Ethical Considerations

No real patient data. All evaluations in this work are conducted on synthetic tasks or publicly available benchmarks. No protected health information, electronic health records, or real patient data were used at any stage of skill evaluation.

Sandboxed execution. Medical skills are evaluated in isolated sandbox environments with no connection to clinical systems. No skill execution produces outputs that could affect real patient care or clinical decision-making.

Non-clinical intent. The AI-generated evaluation scores are research outputs. They are not

intended for, and should not be used in, clinical decision-making without thorough validation by qualified medical professionals.

LLM usage disclosure. GPT-5.1 was used as the primary model for task generation (Step 0) and LLM-as-judge scoring. Claude Sonnet 4.5 was used in backup evaluation runs. All model outputs were treated as intermediate research outputs subject to human oversight.

References

- Aman Arora and 1 others. 2025. [HealthBench: Evaluating large language models towards improved human health](#).
- Yaotian Fang and 1 others. 2025. [PLAY2PROMPT: Zero-shot tool instruction optimization for LLM agents via tool play](#). In *Findings of the Association for Computational Linguistics: ACL 2025*.
- Siyuan Guo and 1 others. 2026. [Learning to rewrite tool descriptions for reliable LLM-agent tool use](#).
- Sheng He and 1 others. 2025. [MedAgentsBench: Benchmarking thinking models and agent frameworks for complex medical reasoning](#).
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Kexin Huang, Serena Zhang, Hanchen Wang, and 1 others. 2025. [Biomni: A general-purpose biomedical AI agent](#). BioRxiv preprint.
- Yixing Jiang and 1 others. 2025. A virtual EHR environment to benchmark medical LLM agents. *NEJM AI*. MedAgentBench.
- Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. 2021. What disease does this patient have? A large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421.
- Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William W. Cohen, and Xinghua Lu. 2019. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577.
- Qiao Jin and 1 others. 2025. [STELLA: Self-evolving LLM agent for biomedical research](#).
- Jon M. Laurent, Albert Bou, Michael Pieler, and 1 others. 2026. [LABBench2: An improved benchmark for AI systems performing biology research](#).

- Jon M. Laurent and 1 others. 2024. [LAB-Bench: Measuring capabilities of language models for biology research.](#)
- Zhehao Li, Junzhi Ning, and 1 others. 2026. [Skills-Bench: Benchmarking how well agent skills work across diverse tasks.](#)
- Zekun Liang and 1 others. 2026. [SkillNet: Create, evaluate, and connect AI skills.](#)
- Yusheng Liao, Shuyang Jiang, Yanfeng Wang, and Yu Wang. 2025. [ReflecTool: Towards reflection-aware tool-augmented clinical agents.](#) In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. Introduces ClinicalAgentBench (CAB).
- MedAgentGym Team. 2025. MedAgentGym: A medical agent training environment. Medical agent training environment.
- Ankit Pal, Logesh Kumar Umapathi, and Malaikannan Sankarasubbu. 2022. MedMCQA: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *Proceedings of the Conference on Health, Inference, and Learning (CHIL)*.
- Long Phan and 1 others. 2025. [Humanity’s last exam.](#) Published in Nature, 2026.
- tau-bench Team. 2024. tau-bench: A benchmark for tool-augmented LLMs. Tool-augmented LLM benchmark.
- ToolSandbox Team. 2024. ToolSandbox: A stateful, conversational, interactive evaluation framework for LLM tool use capabilities. Stateful tool-use evaluation framework.
- David Vilares and Carlos Gómez-Rodríguez. 2019. HEAD-QA: A healthcare dataset for complex reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 960–966.
- Siyu Yuan, Kaitao Song, Jiangjie Chen, Xu Tan, Yongliang Shen, Ren Kan, Dongsheng Li, and Deqing Yang. 2025. [EASYTOOL: Enhancing LLM-based agents with concise tool instruction.](#) In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.

A Evaluation Dimensions: Detailed Definitions

The MED-SKILLBENCH evaluation framework scores each skill–task trial along eight binary dimensions. The first three are designated *core* dimensions; a skill passes overall only if all three core dimensions score 1. The remaining five are *auxiliary* dimensions that capture execution quality beyond the minimum pass threshold. Below we provide the full definition and scoring criteria for each dimension.

1. skill_invoked (Core). The agent explicitly references, reads, or attempts to use the skill specification file during the execution trajectory. A score of 1 is assigned if the agent’s trajectory contains evidence that the skill specification was consulted—for example, a file-read action targeting the skill specification, direct quotation of skill instructions, or an explicit statement acknowledging the skill. A score of 0 indicates the agent completed the task without any evidence of engaging with the provided skill.

2. skill_used_correctly (Core). The agent applies the skill in a manner consistent with the specification’s intended use case, input–output contract, and procedural instructions. A score of 1 requires that the agent’s actions align with the skill’s documented workflow—correct parameter usage, appropriate invocation context, and adherence to any constraints or prerequisites stated in the specification. Partial or superficial references to the skill that do not translate into specification-aligned actions receive a score of 0.

3. task_completion (Core). The agent produces a final output that satisfies the task objective. Scoring considers whether the output addresses the stated goal, provides a substantive answer or artifact, and is not truncated, empty, or obviously incorrect. A score of 1 does not require perfect accuracy but demands that the output constitutes a reasonable, complete attempt.

4. tool_execution_success (Auxiliary). At least one tool invocation during the trajectory produces a verified, non-trivial output. This dimension captures whether the agent successfully grounds the skill specification into executable tool calls that return meaningful results. A score of 0 indicates that all tool calls either failed, produced empty

outputs, or were never attempted despite the skill specifying tool-based workflows.

5. output_quality (Auxiliary). The final output demonstrates domain-appropriate quality: accurate terminology, logically coherent reasoning, and sufficient depth for the task’s complexity. The judge assesses whether the output would be considered adequate by a domain practitioner, accounting for the difficulty of the generated task.

6. reasoning_quality (Auxiliary). The agent’s reasoning trajectory—including intermediate steps, tool-call rationale, and error recovery—demonstrates logical coherence and appropriate problem decomposition. A score of 1 indicates that the agent’s chain of thought is traceable, non-circular, and reflects genuine engagement with the problem rather than superficial pattern matching.

7. efficiency (Auxiliary). The agent completes the task within a reasonable number of iterations relative to task complexity, without excessive redundant actions, repeated failures on the same step, or unnecessary detours. A score of 0 is assigned when the agent exhausts the iteration budget, enters retry loops without progress, or performs substantially more actions than necessary.

8. trajectory_quality (Auxiliary). The overall trajectory exhibits purposeful progression toward the goal, appropriate error handling, and coherent state management across iterations. This holistic dimension captures trajectory-level phenomena not addressed by individual dimensions: does the agent recover from failures constructively, maintain context across steps, and converge toward a solution?

Aggregation. A skill achieves *pass* status when all three core dimensions score 1. A skill achieves *gold* status when all eight dimensions score 1. *Silver* status denotes core pass with at least one auxiliary failure. The RBU gap is computed as the difference between *skill_invoked* and *task_completion* (see Equation 6 in the main text).

B Mandatory Specification-Access Instruction

Every skill evaluation trial injects the following system-level instruction before the task description to ensure the agent engages with the provided

skill specification. This prompt is appended to the agent’s system message and is not visible to the judge.

IMPORTANT: Before attempting this task, you MUST consult the provided skill specification document. This document contains critical instructions, tool specifications, and domain knowledge required to complete the task correctly.

Steps:

1. Review the full skill specification before taking any action.
2. Follow the procedures, constraints, and tool invocations described in the specification.
3. If the specification identifies particular tools, libraries, or commands, use them as directed.
4. Only after reviewing and understanding the specification should you proceed to address the task.

Do NOT skip the specification. Do NOT rely solely on your internal knowledge if the specification provides specific procedures.

This prompt achieves a 96.2% invocation rate across the full corpus of 1,462 skills, validating its effectiveness at directing agent attention. The remaining 3.8% of non-invocation cases arise primarily from skills whose specification documents are empty, malformed, or contain only metadata headers without actionable content.

C Skill Domain Distribution

The 917 Tier-A skills in MED-SKILLBENCH span multiple medical and biomedical domains, reflecting the breadth of open-source medical agent tool repositories. Domain annotations are produced by an automated classifier with manual verification of borderline cases.

Primary domains. The corpus covers the following major areas:

- **Biology & Bioinformatics** — Sequence analysis (BLAST, multiple sequence alignment), gene expression analysis, phylogenetics, genome annotation, protein structure prediction, and pathway analysis tools.
- **Clinical Medicine** — Clinical decision support, diagnostic reasoning guides, treatment protocol references, drug interaction checkers, and clinical calculator wrappers.
- **Drug Development & Pharmacology** — Molecular docking, ADMET prediction, compound screening, pharmacokinetic modeling, and drug–target interaction databases.
- **Chemoinformatics** — Chemical structure manipulation (RDKit-based), molecular fingerprinting, SMILES/InChI conversion, and chemical property prediction.

- **Proteomics & Structural Biology** — Protein–protein interaction analysis, mass spectrometry data processing, structural alignment, and binding site prediction.
- **Genomics & Variant Analysis** — Variant calling, VCF processing, genome-wide association analysis, and annotation pipelines.
- **Medical Literature & Evidence** — PubMed search, systematic review automation, evidence grading, and citation management.
- **Laboratory & Safety** — Laboratory protocol guides, biosafety references, equipment operation procedures, and regulatory compliance checklists.
- **Medical Imaging** — DICOM processing, radiological analysis aids, and image segmentation tool wrappers.
- **Public Health & Epidemiology** — Epidemiological modeling, outbreak analysis, surveillance data processing, and population health statistics.

Distribution characteristics. The domain distribution is intentionally unbalanced, reflecting the natural distribution of skills in the source repositories. Biology & Bioinformatics and Clinical Medicine together account for the largest share, followed by Drug Development and Medical Literature tools. This imbalance is preserved rather than artificially balanced, as it reflects the real-world landscape of available medical agent tools.

D Full Benchmark Statistics

Table 10 extends the summary in Table 7 with additional columns for absolute match counts, unique skill counts, and the top match types observed per benchmark.

Notes on matched counts. The “Matched” column reports unique questions with at least one skill assignment; a single question may be matched to multiple skills, yielding approximately 3,898 total skill–question pairs across the 2,596 matched questions. Relevance annotations indicate that 91.5% of matches are classified as *weak* relevance (the skill provides tangential support) and 8.5% as *strong* relevance (the skill directly addresses the core reasoning required by the question).

Skill concentration analysis. Across all benchmarks, the top three matched skills are agent-browser (appearing in 78% of all matched pairs), ai-analyzer (13.8%), and

Table 10: Extended skill–benchmark matching statistics across 12 medical QA benchmarks (917 Tier-A skills). “Matched” indicates the number of questions with at least one skill match. “Top match types” lists the two most frequent match type annotations for each benchmark.

Benchmark	Total Q	Matched	Match %	Skills	Top Match Type 1	Top Match Type 2
Lab-Bench	900	416	46.2	90	database_query	literature_search
MedAgents	332	96	28.9	16	literature_search	reasoning_guide
MedMCQA	4,183	1,071	25.6	50	literature_search	database_query
PubMedQA	1,000	194	19.4	35	literature_search	database_query
HeadQA	2,742	499	18.2	66	literature_search	reasoning_guide
SGI-Reasoning	72	9	12.5	17	reasoning_guide	literature_search
MMLU-Med	1,089	127	11.7	42	literature_search	reasoning_guide
MedQA-USMLE	1,273	71	5.6	21	literature_search	reasoning_guide
LabSafety	765	42	5.5	3	literature_search	reasoning_guide
MedCalc-Bench	1,067	47	4.4	5	database_query	reasoning_guide
MedBullets-5op	298	15	5.0	6	literature_search	reasoning_guide
MedBullets-4op	298	8	2.7	6	literature_search	reasoning_guide
Overall	14,019	2,596	18.5	178	literature_search (71.7%)	database_query (21.0%)

Table 11: Evaluation results by processing batch, demonstrating stability across the corpus.

Batch	n	Pass %	Gold %	RBU Gap
1 (1–400)	400	57.8	15.0	0.333
2 (401–800)	400	65.5	23.8	0.173
3 (801–1200)	400	68.3	22.8	0.168
4 (1201–1462)	262	67.2	20.2	0.199
Overall	1,462	64.4	20.5	0.220

bio-entrez-search (12.2%). This heavy concentration underscores that current medical skill repositories favor general-purpose retrieval and analysis tools over domain-specialized reasoning aids. Lab-Bench is an exception, with 90 unique skills matched—the highest skill diversity of any benchmark—consistent with its emphasis on specific laboratory protocols and bioinformatics pipelines.

E Evaluation Stability

To assess evaluation stability across the corpus, we report results stratified by processing batch in Table 11. Skills were processed in sequential batches of 400 (with the final batch containing the remaining 262 skills).

Batches 2–4 show stable performance, with pass rates in the 65.5–68.3% range and RBU gaps between 0.168 and 0.199, confirming that the evaluation framework yields consistent results across the corpus.

F Skill-Augmented QA: Detailed Results

This appendix provides extended results for the skill-augmented QA evaluation reported in Sec-

tion 4.6.

Per-benchmark breakdown. Table 12 reports per-benchmark accuracy under both conditions for all evaluated models, including benchmarks omitted from the main-text table due to small matched sample sizes.

Statistical details. For each benchmark–model pair, we report the McNemar χ^2 statistic and two-sided p -value. Bootstrap 95% confidence intervals for the uplift are computed via 10,000 resamples of the paired (with, without) binary outcome vectors. [TODO: Populate statistical results once Step 2 experiments complete.]

Uplift distribution. Per-question uplift takes one of three values: +1 (with-skill correct, without-skill incorrect), 0 (same outcome), or −1 (with-skill incorrect, without-skill correct). The proportion of +1 vs. −1 outcomes determines the direction and magnitude of the aggregate effect. [TODO: Report uplift distribution breakdown and add histogram figure.]

G SGI Sensitivity Analysis

We verify that the core findings are robust to the choice of w_c in the SGI definition (Equation (10)). The dominance principle (Equation (12)) requires $w_c > 0.75$ to guarantee that any core-failing skill ranks below any core-passing skill. We report results for $w_c \in \{0.60, 0.70, 0.75, 0.80\}$.

[TODO: Compute SGI under all four w_c values from Step 0 data and verify that domain rankings are stable (Kendall’s $\tau > 0.9$).] At $w_c = 0.60$, the dominance property is violated: a core-failing skill

Table 12: Extended skill-augmented QA results across all 12 medical benchmarks. n : number of evaluated (question, skill) pairs per benchmark. Bold Δ indicates $p < 0.05$ (McNemar’s test).

Benchmark	n	GPT-5.1			Claude Sonnet 4.5		
		w/o	w/	Δ	w/o	w/	Δ
Lab-Bench	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MedAgents	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MedMCQA	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
PubMedQA	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
HeadQA	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
SGI-Reasoning	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MMLU-Med	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MedQA-USMLE	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
LabSafety	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MedCalc-Bench	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MedBullets-5op	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MedBullets-4op	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Overall	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 13: SGI sensitivity to w_c . Domain rankings and the separation between core-passing and core-failing skills are reported for each weight.

Metric	$w_c=0.60$	$w_c=0.70$	$w_c=0.75$	$w_c=0.80$
Mean SGI (all)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Mean SGI (core-pass)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Mean SGI (core-fail)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Separation (pass – fail)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Domain rank correlation (τ)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

We reject H_0 at $\alpha = 0.05$ if $\chi^2_{\text{McN}} > 3.841$.

Multi-match dependence. When a question matches multiple skills ($|\mu(x_i)| > 1$), pairs sharing the same question have correlated without-skill outcomes, violating the independence assumption. We address this via clustered bootstrap (clustering by question index i) as a robustness check alongside the standard McNemar test.

Bootstrap confidence intervals. For Δ_{uplift} , we construct bootstrap 95% CIs by resampling the paired outcome vectors with replacement $B = 10,000$ times, clustering by question index i :

$$\text{CI}_{95\%}(\Delta_{\text{uplift}}) = [\hat{\Delta}^{*(0.025)}, \hat{\Delta}^{*(0.975)}], \quad (23)$$

where $\hat{\Delta}^{*(\alpha)}$ is the α -quantile of the bootstrap distribution.

Groundability-uplift correlation. The aggregate interface-type correlation (Equation (19)) is computed over $|\mathcal{I}| = 5$ data points. With $n = 5$, the critical value for significance at $\alpha = 0.05$ is $|r| > 0.878$, yielding negligible statistical power. We therefore use the per-skill SGI→uplift regression (Equation (20)) as the primary test, which operates over individual skills (higher n) and provides substantially greater power. [TODO: Report both r_{GU} and $\hat{\beta}_1$ with standard errors once Step 2 experiments complete.]

with perfect auxiliary quality achieves SGI = 0.80, outscoring a core-passing skill with zero auxiliary quality (SGI = 0.60). At $w_c \geq 0.75$, the separation is guaranteed.

H Statistical Testing Details

This appendix provides the full statistical testing framework for the skill-augmented QA evaluation.

McNemar’s test for paired binary outcomes. For each skill–question pair $(x_i, s, g_i) \in \mathcal{P}_\mu$, let $c_{i,s}^+ = \mathbf{1}[\mathcal{M}(x_i, s) = g_i]$ (with-skill correctness) and $c_{i,s}^- = \mathbf{1}[\mathcal{M}(x_i, \emptyset) = g_i]$ (without-skill correctness). The discordant counts are:

$$\begin{aligned} b &= \sum_{(i,s)} \mathbf{1}[c_{i,s}^+ = 1 \wedge c_{i,s}^- = 0], \\ c &= \sum_{(i,s)} \mathbf{1}[c_{i,s}^+ = 0 \wedge c_{i,s}^- = 1]. \end{aligned} \quad (21)$$

Under H_0 : skill availability has no effect, b and c follow a binomial distribution with equal probabilities. The McNemar statistic is:

$$\chi^2_{\text{McN}} = \frac{(b - c)^2}{b + c} \sim \chi^2(1). \quad (22)$$